APO-TRIFLUOPERAZINE
Trifluoperazine Tablets BP 1mg, 5mg
Antipsychotic- Antiemetic- Antianxiety

Pharmacodynamics
Trifluoperazine, a piperazine phenothiazine derivative, has pharmacologic effects similar to those of chlorpromazine. Trifluoperazine has greater psycho-pharmacologic potency and a more prolonged duration of action than chlorpromazine.
Trifluoperazine has weak anticholinergic and sedative effects, strong extrapyramidal effects and strong antiemetic activity.
Phenothiazines are generally well absorbed from the gastrointestinal tract and from parenteral sites; however, absorption may be erratic, particularly following oral administration. There are considerable inter-individual variations in peak plasma concentrations. Therapeutic ranges and the relationship of plasma drug concentration to clinical response and toxicity have not been clearly established. Phenothiazines are highly protein bound.
Phenothiazines are extensively metabolized, principally in the liver. Most metabolites are pharmacologically inactive. Phenothiazines and metabolites are excreted in urine and feces. They are excreted in feces via biliary elimination principally as metabolites and also appear to undergo enterohepatic circulation.
Phenothiazines readily cross the placenta. It is not known if the drugs are distributed into breast milk.

Indication
For the treatment of psychotic disorders particularly to produce a quieting effect in hyperactive or excited psychotic patients; for the treatment of behavioural disorders in adults, in the aged and children, and to prevent or relieve nausea and vomiting due to stimulation of the chemoreceptor trigger zone.

Dosage
Adjust dosage to the individual and severity of condition. When maximum therapeutic response is obtained, dosage is gradually reduced to maintenance level.

Oral: To control anxiety, for adults, treatment normally starts with 1mg twice daily. Doses exceeding 4mg daily are rarely needed.
Psychotic patients: for adequately supervised adults, the usual starting dose ranges between 2 to 5mg twice daily with optimum response occurring in most patients on 15 to 20mg daily, but some patients may require as much as 40mg daily. Ordinarily 2 to 3 weeks are required to obtain optimum response and improvement. Maintenance dosage should not be instituted prematurely. In hospitalized psychotic children: 6 to 12 years old, 1mg may be given once or twice daily. This dose may be increased if necessary but extra caution must be exercised at higher doses. As an antiemetic: for adults, 1 or 2mg twice daily, or as required.

Contraindication
Comatose or depressed states due to CNS depressants; blood dyscrasias; bone marrow depression; marked liver damage. Hypersensitivity to trifluoperazine; cross-sensitivity to other phenothiazines may occur.

Warning and Precautions
Hypotension, which is typically orthostatic, may occur especially in elderly and in alcoholic patients. This effect may be additive with other agents that cause a lowering of blood pressure. Exercise special care in those patients in whom a hypotensive crisis would be undesirable such as those with arteriosclerosis or other cardiovascular disease.
Prolongation of the QT interval, flattening and inversion of the T wave and appearance of a wave tentatively identified as a bifid T or a U wave have been observed in some patients receiving phenothiazines. These changes appear to be reversible and may resemble quinidine-like alterations or those induced by hypokalemia. Give phenothiazines cautiously to patients with heart disease. A baseline ECG is suggested for older patients.
Unexplained and sudden deaths have been reported in hospitalized psychotic patients receiving phenothiazines. In some, myocardial lesions have been observed. Previous brain damage or seizures may also be predisposing factors; high doses should be avoided in known seizure patients. Several patients have shown sudden exacerbations of psychotic behavior patterns shortly before death.
Most reported cases of agranulocytosis associated with the administration of phenothiazine derivatives have occurred between the fourth and tenth week of treatment. Therefore, observe patients on prolonged therapy with particular care during that time for the appearance of such signs as sore throat, fever and weakness. If these symptoms appear, discontinue the drug and perform WBC and differential counts. Routine periodic WBC and differential counts are also advisable during prolonged but uncomplicated therapy.
If bilirubinemia, bilirubinuria or icterus occur, discontinue the drug and perform liver function tests. Routine periodic liver function tests are advisable during prolonged therapy.
Phenothiazines have been associated with retinopathy. Discontinue trifluoperazine if retinal changes are observed.
Use trifluoperazine cautiously in patients with a history of seizures.
The occasional increase in physical activity resulting from trifluoperazine administration may augment the severity of pain in angina pectoris patients. Observe such patients carefully and withdraw the drug if necessary.
Pregnancy: Teratogenic effects have not been demonstrated. It is generally recognized that caution should always be observed when prescribing for the pregnant patient, especially during the first trimester and near term. Trifluoperazine should not be used during pregnancy unless the physician considers that the benefit outweighs the possible risk.
Lactation: There is evidence that phenothiazines are excreted in the milk of nursing mothers. Use with caution because of the possible sedative and anticholinergic side effects to the infant.
Occupational hazards: Where patients are participating in activities requiring complete mental alertness such as driving an automobile or operating machinery, administer the phenothiazine cautiously, forewarn the patient and increase the dosage gradually.
Trifluoperazine's antiemetic effect may mask signs of overdosage of toxic drugs or obscure the diagnosis of conditions such as intestinal obstruction and brain tumor.
Geriatrics: Use in reduced dose. Many of the conditions, particularly cardiovascular, that it may affect adversely are common in the aged.

Drug Interactions:
Phenothiazines may increase the effects of general anesthetics, opiates, barbiturates, alcohol and other CNS depressants as well as atropine and phosphorus insecticides.
If agents such as sedatives, narcotics, anesthetics, tranquilizers or alcohol are used either simultaneously or successively with trifluoperazine, the possibility of an undesirable additive depressant effect should be considered.
Avoid epinephrine in the treatment of phenothiazine induced hypotension because phenothiazines may reverse epinephrine's action and thereby cause a further fall in blood pressure.
Phenothiazines may lower the convulsive threshold; dosage adjustment of anticonvulsants may be necessary. Potentiation of anticonvulsant effects does not occur. However, it has been reported that phenothiazines may interfere with the metabolism of phenytoin and thus precipitate phenytoin toxicity.
Drugs which lower the seizure threshold, including phenothiazine derivatives, should not be used with metrizamide. As with other phenothiazine derivatives, trifluoperazine should be discontinued at least 48 hours before myelography, should not be resumed for at least 24 hours post procedure, and should not be used for the control of nausea and vomiting occurring either prior to myelography or post procedure.
Phenothiazines may diminish the effect of oral anticoagulants.
Concomitant administration of propranolol with phenothiazines results in increased plasma levels of both drugs.
Lithium: Although most patients receiving lithium and a phenothiazine antipsychotic agent concurrently do not develop unusual adverse effects, an acute encephalopathic syndrome occasionally has occurred, especially when high serum lithium concentrations were present. Patients receiving such combined therapy should be observed for evidence of adverse neurologic effects; treatment should be promptly discontinued if such signs or symptoms appear.

Pregnancy and Lactation
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Lactation: There is evidence that phenothiazines are excreted in the milk of nursing mothers. Use with caution because of the possible sedative and anticholinergic side effects to the infant.

Side Effects

Adverse effects with different phenothiazines vary in type, frequency and mechanism of occurrence, i.e., some are dose-related, while others involve individual patient sensitivity. Some adverse effects may be more likely to occur, or occur with greater intensity, in patients with special medical problems, e.g., patients with mitral insufficiency or pheochromocytoma have experienced severe hypotension following recommended doses of certain phenothiazines.

In general, members of the piperazine group of phenothiazines have more marked stimulating effects, are more likely to cause motor disorders associated with extrapyramidal reactions, particularly in children but are less likely to cause blood dyscrasias, hypotension, tachycardia, and drowsiness than the members of the other phenothiazine groups.

Not all of the following adverse reactions have been observed with every phenothiazine derivative, but they have been reported with one or more and should be borne in mind when drugs of this class are administered. Behavioral reactions: oversedation; impaired psychomotor function; paradoxical effects, such as agitation, excitement, insomnia, bizarre dreams, aggravation of psychotic symptoms; and toxic confusional states. CNS: extrapyramidal reactions, including Parkinsonism (with motor retardation, rigidity, masklike facies, tremor, salivation, etc.); dystonic reactions (including facial grimacing, tics, torticollis, oculogyric crises, etc.); and akathisia. Persistent dyskinesias resistant to treatment have also been reported. In addition, slowing of EEG, disturbed body temperature, and lowering of the convulsive threshold have occurred. Tardive dyskinesias: Tardive dyskinesia may appear in some patients on long-term antipsychotic therapy or after it has been discontinued.

The risk appears to be greater in elderly patients on high dose therapy, especially females. The symptoms are persistent and in some patients appear to be irreversible. The syndrome is characterized by rhythmical involuntary movements of the tongue, face, mouth or jaw (e.g. protrusion of tongue, puffing of cheeks, puckering of mouth, chewing movements). Sometimes these may be accompanied by involuntary movements of extremities.

There is no known effective treatment for tardive dyskinesia; antiparkinsonian agents usually do not alleviate the symptoms. All antipsychotic agents should be discontinued if these symptoms appear. Should it be necessary to reinstitute treatment, or increase the dosage of the agent, or switch to a different antipsychotic agent, the syndrome may be masked. The physician may reduce the risk of this syndrome by minimizing the unnecessary use of neuroleptics and reducing the dose or discontinuing the drug. If possible, when manifestations of this syndrome are recognized, particularly in patients over the age of 50. Fine vermicular movements of the tongue may be an early sign of the syndrome. If the medication is stopped at that time, the syndrome may not develop.

Autonomic nervous system: dry mouth, fainting, stuffy nose, photophobia, blurred vision, miosis. The drug may be sufficiently antiadrenergic in large doses to give symptoms of blockade, for e.g. retrograde ejaculation.

Gastrointestinal: anorexia, increased appetite, gastric irritation, nausea, vomiting, constipation, paralytic ileus.

Endocrine system: excess prolactin secretion, altered libido, menstrual irregularities, lactation, false positive pregnancy tests, inhibition of ejaculation, gynecomastia, weight gain.

Skin: itching, rash, hypertrophic papillae of the tongue, angioneurotic edema, erythema, allergic purpura, exfollative dermatitis, contact dermatitis.

Cardiovascular effects: hypotension, tachycardia, ECG changes (see Precautions).

Blood dyscrasias: agranulocytosis, leukopenia, granulocytopenia, eosinophilia, thrombocytopenia, anemia, aplastic anemia, pancytopenia.

Allergic reactions: fever, laryngeal edema, angioneurotic edema, asthma.

Hepatic: jaundice, biliary stasis.

Urinary disturbances: retention, incontinence.

Abnormal pigmentation: more recently, a pecullar skin-eye syndrome has been recognized as an adverse effect following long-term treatment with phenothiazines. This reaction is marked by progressive pigmentation of areas of skin or conjunctiva and /or discoloration of the exposed sclera and cornea. Opacities of the anterior lens and cornea described as irregular or stellate in shape have also been reported. Patients receiving higher doses of phenothiazines for prolonged periods should have periodic complete eye examinations.

Miscellaneous: Silent pneumonias including aspiration may occur.

Overdose

Symptoms: Drowsiness, confusion, disorientation, followed in more severe cases by coma and areflexia. Dry mouth, nasal congestion, blurred vision, urinary retention, constipation. Postural hypotension which may be early in onset, severe, and persistent; tachycardia, cardiac arrhythmias, cardiovascular collapse. Respiratory depression may be a late manifestation of severe phenothiazine intoxication. Motor restlessness, hyperreflexia, and convulsions. Hypothermia or hyperthermia.

Note: Acute extrapyramidal symptoms, such as dystonia and oculogyric crisis, have not been reported with acute thioridazine overdosage, but are possible.

Treatment: There is no specific antidote for acute phenothiazine poisoning. Therefore, treatment is directed at minimizing the amount of drug absorbed, eliminating the absorbed drug from the body, and combating the toxic effects of the overdose.

Elimination of the offending drug:

1) Emesis: If the patient is conscious, induce vomiting mechanically or with emetics.

2) Perform gastric lavage if the pharyngeal and laryngeal reflexes are present, and if less than 4 hours have elapsed since ingestion. Do not attempt gastric lavage on an unconscious patient unless cuffed endotracheal intubation has been performed to prevent aspiration and pulmonary complications.

3) Catharsis: Following gastric lavage, a saline cathartic [sodium or magnesium sulfate 30g (1oz.) plus charcoal 50 to 100g in 250ml (1 cup) of water] may be introduced and left in the stomach. These should be repeated every 4 to 6 hours for 24 hours as the drug is enterohepatically recycled.

4) Maintain pulse, BP and ECG. Perform abdominal x-ray as phenothiazines are radiopaque and intact tablets may be visualized.

Note: Peritoneal dialysis and hemodialysis have been found ineffective in the treatment of acute phenothiazine poisoning. Hemoperfusion may be useful in severe cases.

Maintenance of adequate pulmonary ventilation: Perform pharyngeal and tracheal suction diligently to remove excess mucous secretions. Judicious administration of oxygen is also indicated. However, oxygen without assisted respiration must be used with caution, as its use in hypoventilation hypoxia may result in further respiratory depression and hypercapnia. In more critical cases, endotracheal intubation or tracheotomy, with or without assisted respiration, may be necessary.

Administer fluids and place patient in Trendelenburg position. If responsive, dopamine may be required. Dopamine: 2 to 5µg/kg/min i.v. initially and gradually increased to 50µg/kg/min as required.

Treat seizures with i.v. diazepam.

General supportive measures: Good nursing care is prime importance, particularly in the comatose patient, and should include regular observation and accurate recording of the vital signs and depth of coma, maintenance of a free airway, frequent turning and other routine measures usually adopted with unconscious patients.

Careful supervision and recording of fluid intake and output is essential. Treat hyperthermia with tepid sponging. Do not use alcohol. For hypothermia, keep patient warm but avoid overheating.

Storage Conditions

Protect from light. Store below 30°C.

Supplied

Each blue, round, biconvex, film-coated tablets contains: 1mg or 5mg of trifluoperazine as trifluoperazine HCl. Imprinted “1” or “5” respectively. Available in bottles of 100 tablets.

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